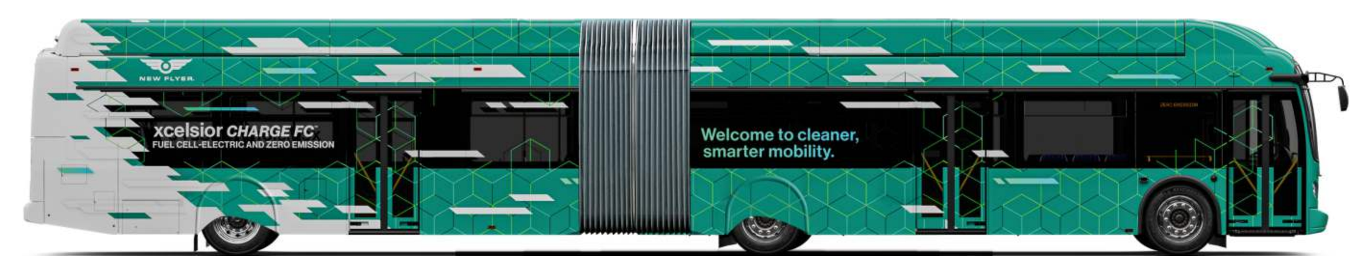
EMERGENCY RESPONSE GUIDE

For a

xcelsior *CHARGE FC*™

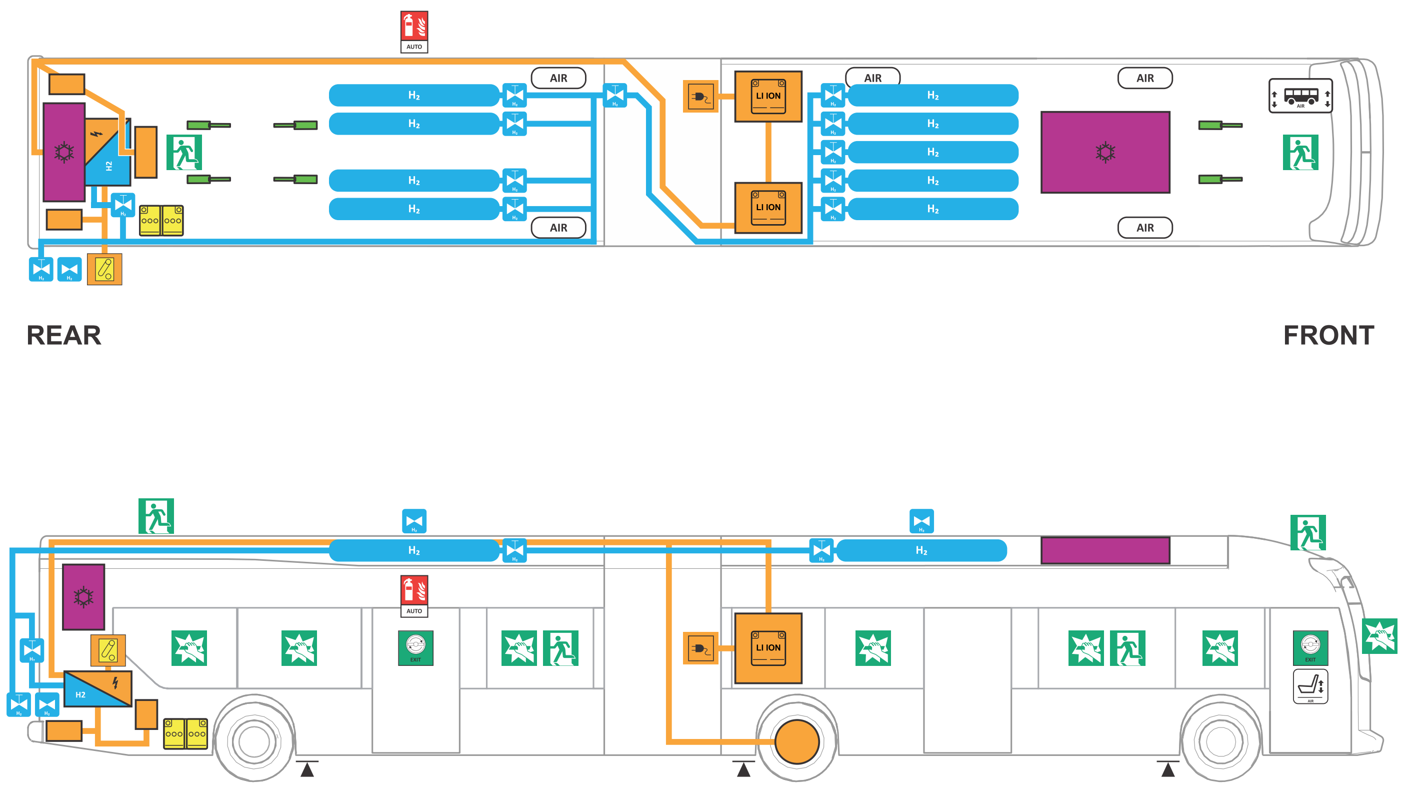
LI ION

HYDROGEN-ELECTRIC
60FT. TRANSIT BUS



Revision A: May 23 2023

0. RESCUE SHEET



REAR FRONT

Fuel Cell Propulsion	Fuel Cell	Manual Gas Shut-Off Valve	Overpressure Safety Valve	Gas Line	Hydrogen Tank	Automatic Fire Suppression
High Voltage Li-ion Battery	High Voltage Component	Plug-in Charge Port	Disconnect High Voltage	High Voltage Power Cable	Air Conditioning	Low Voltage Battery

Preloaded Spring	Emergency Exit	Break To Obtain Access	Emergency Door Opener	Air Tank	Seat Adjustment	Height Adjustment	Lifting Point

1. IDENTIFICATION / RECOGNITION

2. IMMOBILIZATION / STABILIZATION / LIFTING

	Kneel switch located in driver's area
	Seat height adjustment on driver's seat
	Lift points

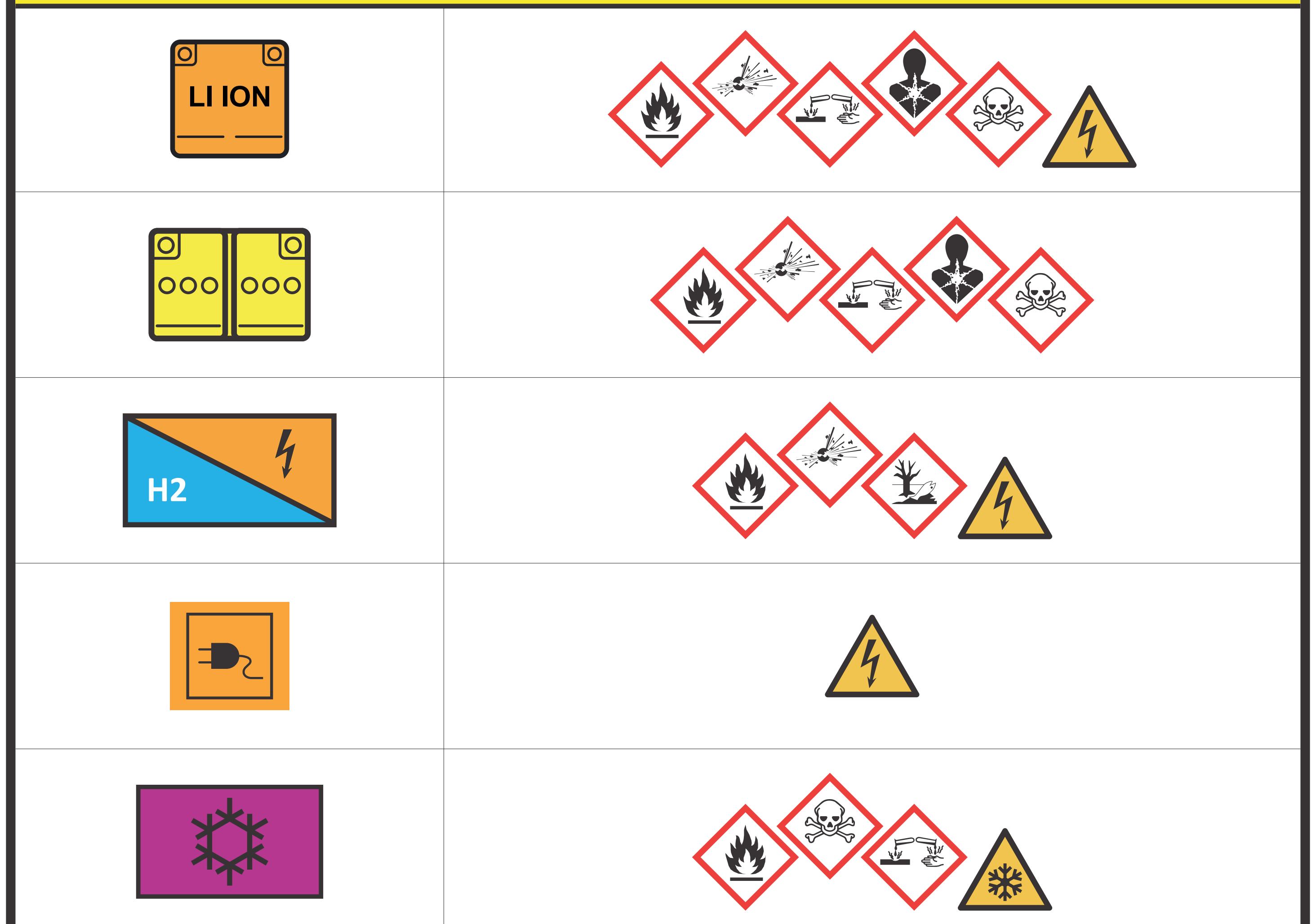
3. DISABLE DIRECT HAZARDS / SAFETY REGULATIONS

	Shut down high voltage

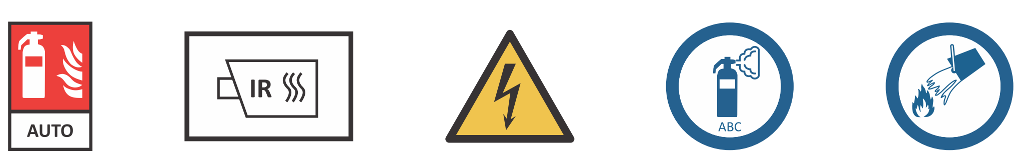
4. ACCESS TO THE OCCUPANTS

	Two exit doors
	Two roof hatch exits, four emergency window exits
	Break windows to obtain access
	Steering wheel tilt control on steering column
	Seat height adjustment on driver's seat
	Longitudinal seat adjustment on driver's seat

5. STORED ENERGY / LIQUIDS / GASES / SOLIDS

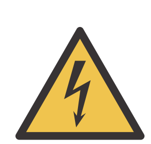


6. IN THE EVENT OF A FIRE



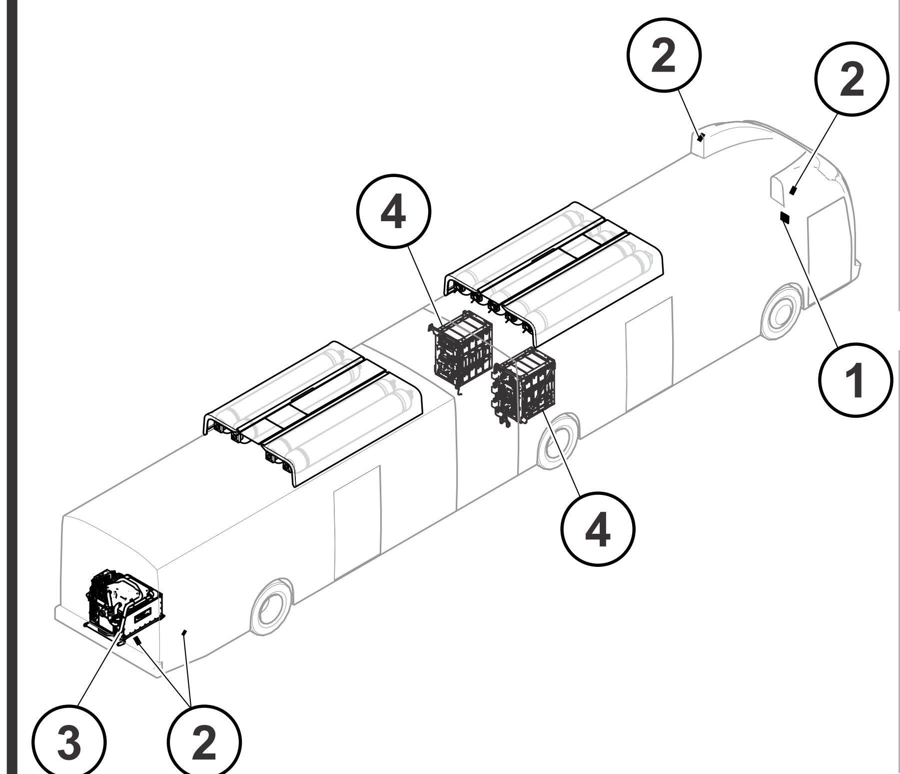
Pressure Relief Device (PRD) relief hydrogen system pressure when the temperature of the hydrogen tank rises above 230°F (110°C).

7. IN CASE OF SUBMERSION



1. IDENTIFICATION / RECOGNITION

The vehicle can primarily be identified as a New Flyer electric vehicle using the VIN decal located overhead, just inside the entrance door area. Alternatively, look for a fill box panel in the curbside access door, energy storage system enclosures as shown, rooftop fuel tanks and fuel cell in the rear propulsion compartment of the vehicle.



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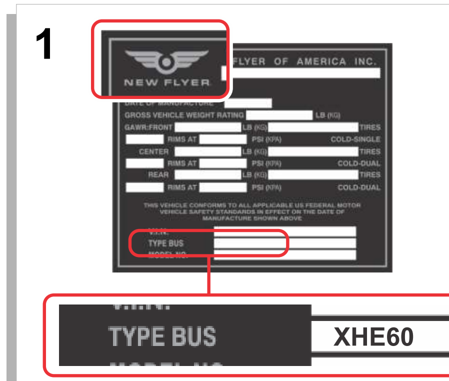
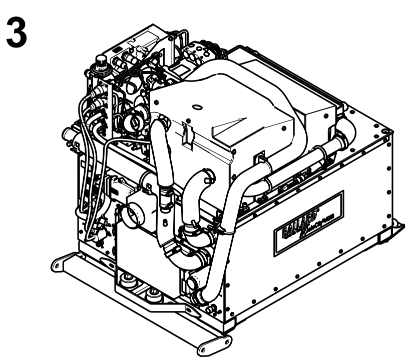


VIN DECAL

2

CH2 DECALS

3



FUEL CELL

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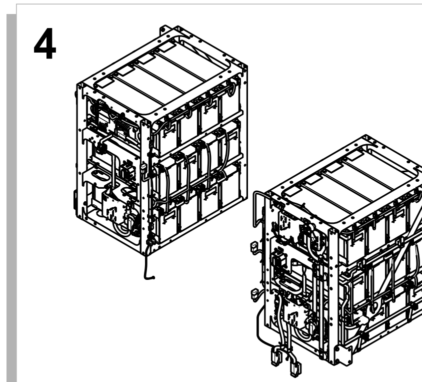
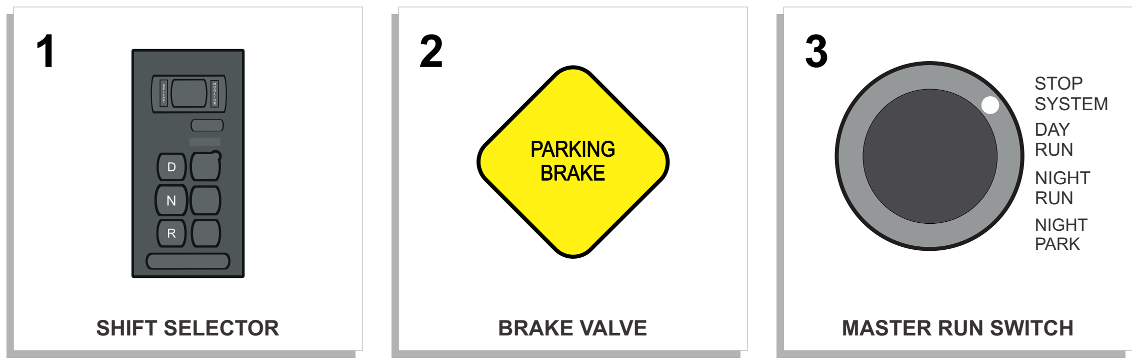


ENERGY STORAGE SYSTEM

2. IMMOBILIZATION / STABILIZATION / LIFTING

1. Shift the vehicle into neutral by pressing N on the shift selector located to the right of the electronic instrument panel.
2. Apply the parking brakes using the brake valve on the driver's side console.
3. Shutdown the vehicle by placing the Master Run switch on the driver's side console to the STOP-SYSTEM position.
4. Chock wheels.



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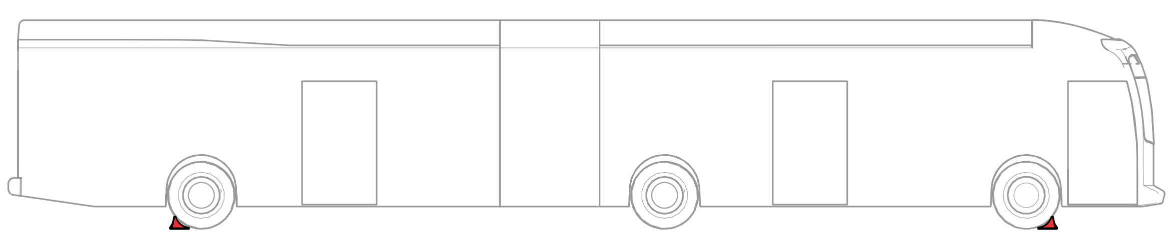
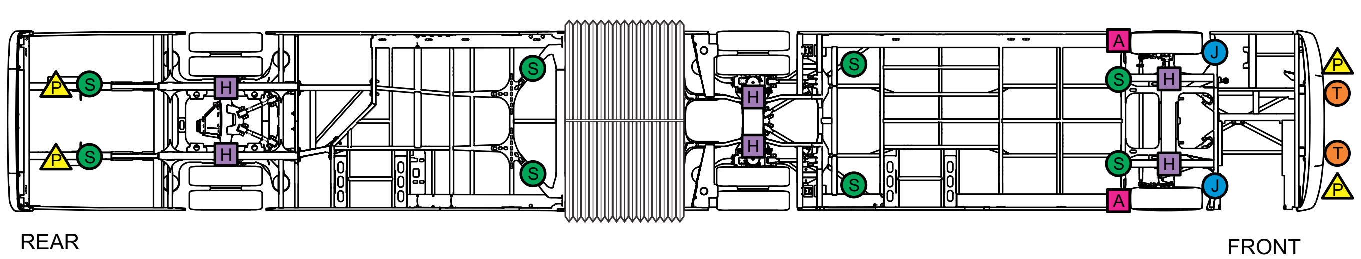


FRONT & REAR WHEEL CHOCKS

In the event of submersion or collision, where vehicle movement may be required for salvage or access purposes, only connect to the points on the frame designated for pulling the vehicle. Jacking and hoisting support can be provided to the vehicle using the locations identified in the following diagram.



LEGEND:

- S - Jack Stand Points
- A - Jacking Adapter/Lifting Locations
- J - Jacking Pads
- P - Pulling/Lifting Points
- T - Towing Adapter Locations
- H - Hoisting Points

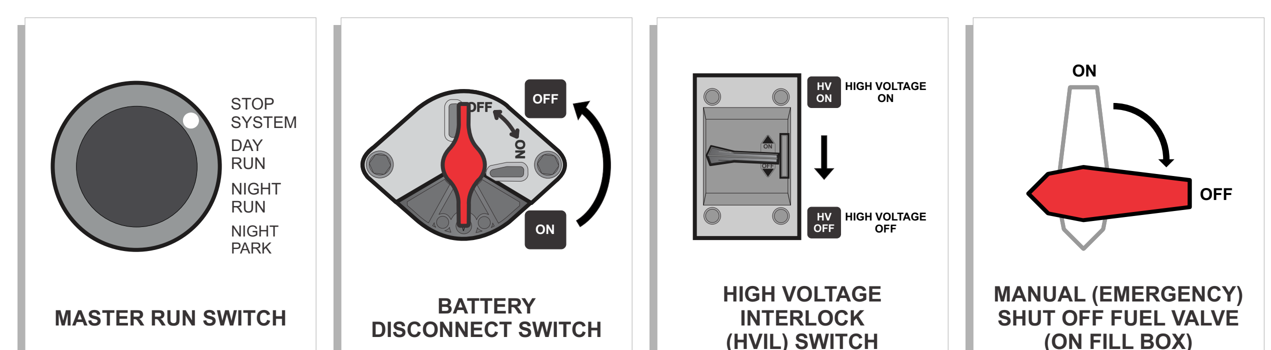
3. DISABLE DIRECT HAZARDS / SAFETY REGULATIONS

DANGER

- Be familiar with and follow applicable National, State/Provincial and Local Fire and Safety Regulations.
- This is an Electric Drive vehicle which contains 600 VDC and 230 VAC high voltage equipment in the rear compartment and on the rooftop of the vehicle.
- High voltage cables can be identified by an orange outer covering. A High Voltage Disconnect switch protects all circuits and components, BUT it is still possible to receive a fatal electric shock from the system!
- Use extreme caution when handling, orange high voltage cables or Electric drive components, as this could result in severe injury or death!
- If working near high voltage cabling or components, use 1,000 VDC electrical gloves, rubber-soled shoes and make sure you and the surrounding area are DRY!

In the event of an emergency, follow the evacuation and shutdown procedure in the sequence shown:

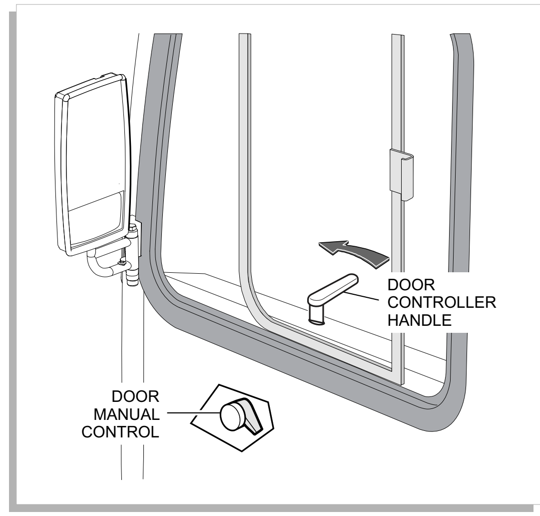
1. Shutdown the vehicle by placing the Master Run switch on the driver's side console to the STOP-SYSTEM position.
2. If accessible, turn the 12/24 VDC Battery Disconnect & High Voltage Interlock switches, located behind the rear curbside access door to the OFF position.
3. If accessible, shut off fuel delivery to the fuel cell by turning the quarter-turn valve, located behind the rear curbside fuel filler access door to the OFF position.



4. ACCESS TO THE OCCUPANTS

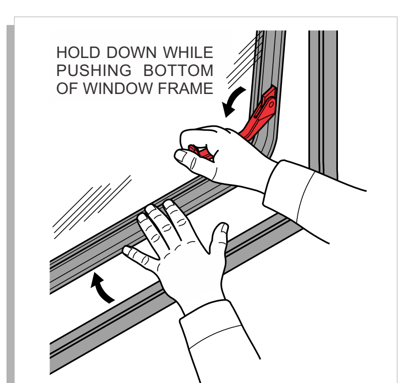
The entrance and exit doors can be opened from the exterior of the vehicle using the controls on the driver's side console.

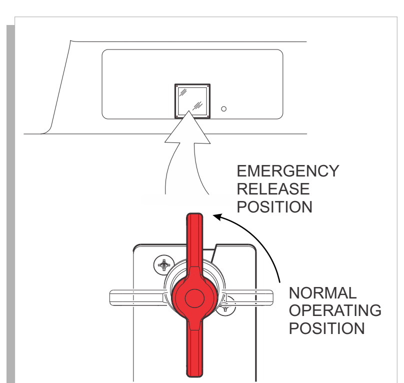
1. **Slide the front portion of the driver's window back to gain access to the door controller handle on the side console.**
2. **Turn the door controller handle to the FRONT, FRONT REAR or REAR FRONT positions to open the entrance door.**
3. **If the entrance door does not open, exhaust air by turning the door manual control valve on the side console to the OFF position. Open the door manually by pulling out the door halves at the seal.**

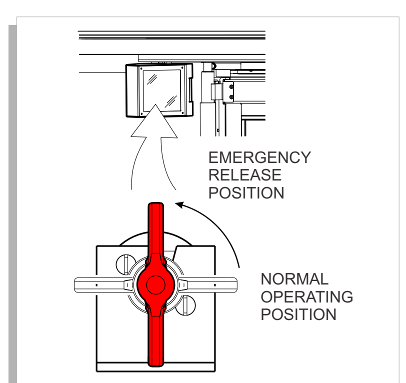


The windows on the bus are breakable and there are also multiple emergency exit windows that can be opened from the inside of the vehicle. Emergency exit windows can be identified by the red release handles on the window frame as well identifying decals. To operate the emergency window, pull the red handle down and hold. Push out on the bottom of the window frame. The window will open on hinges at the top of the frame.

Rotary dump valves are provided at each door. The emergency release control valves are located behind breakable windows. In an emergency, break the glass to access the control valve knob. Rotate the knob 90° counter-clockwise, then push the doors open.



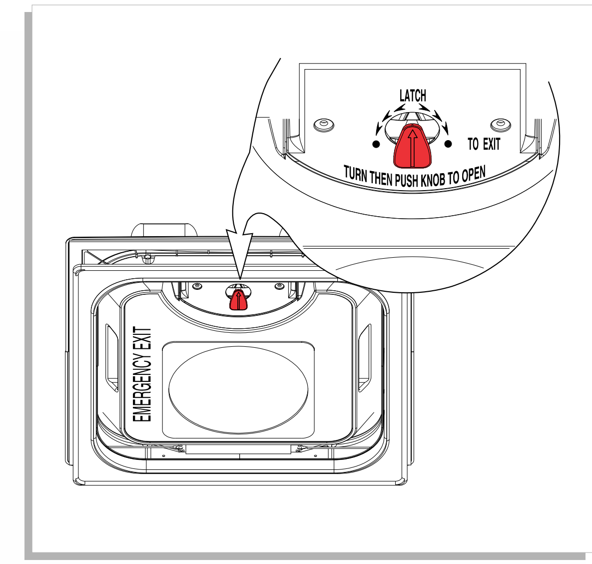




The roof hatches function as emergency exits and are identified by decals on the hatch panel. Proceed as follows to operate the emergency exit:

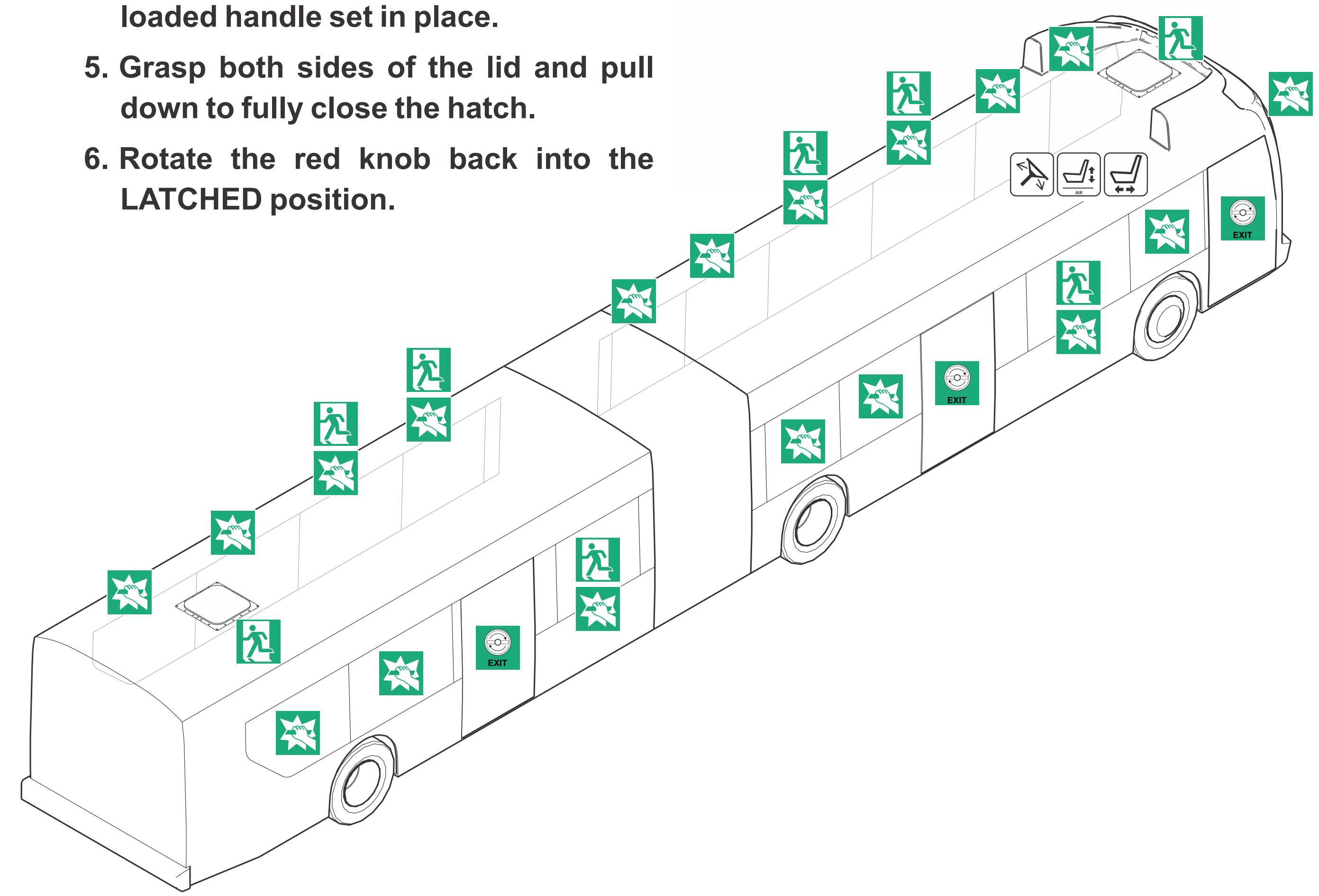
OPENING

1. **Rotate the red knob 90°.**
2. **Push the red knob into the lid.**
3. **Continue to push the lid to the fully OPEN position.**

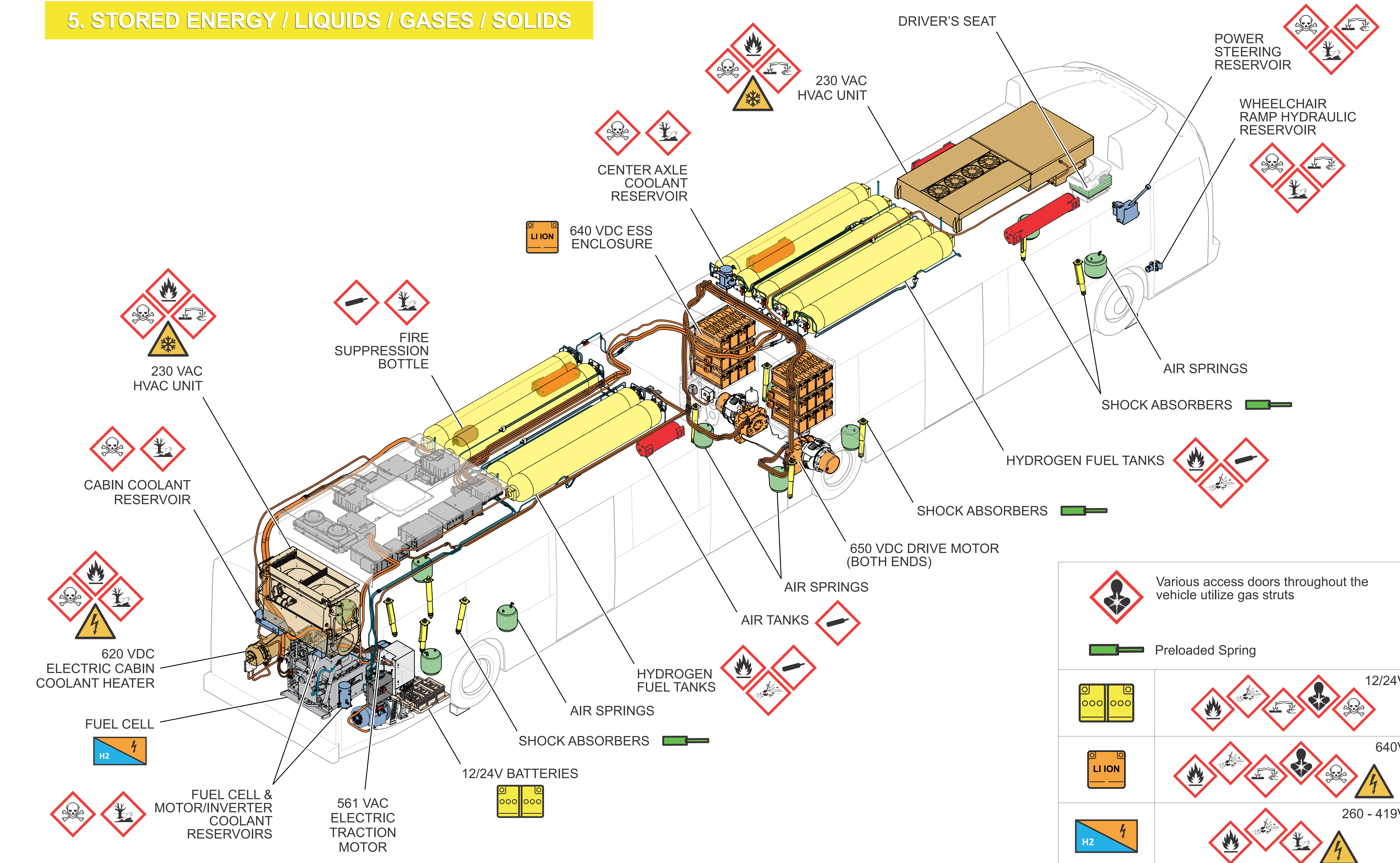


CLOSING

1. **Ensure the release hinge is in the UPWARD position.**
2. **Lower the lid into position.**
3. **Guide the release hinge into the handle base on the lid.**
4. **Pull down on the top of the lid to force the release hinge and the lid together until you hear the spring loaded handle set in place.**
5. **Grasp both sides of the lid and pull down to fully close the hatch.**
6. **Rotate the red knob back into the LATCHED position.**



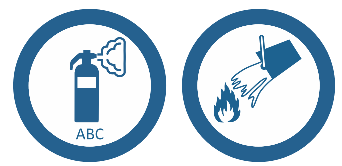
5. STORED ENERGY / LIQUIDS / GASES / SOLIDS



6. IN THE EVENT OF A FIRE

⚠ WARNING

Lithium-ion batteries use a carbon negative electrode and a Lithium Cobalt Manganese Nickel Oxide positive electrode. The electrolyte is a solution of lithium hexaflourophosphate in a mixture of organic solvent. Exposed electrolyte solution is corrosive and may be flammable in sufficient concentration.



- DO NOT USE Class D fire extinguisher.
- USE CLASS A, B or C fire extinguishers for small fires.
- USE WATER in large amounts for large fires.
- Wear full protective clothing including helmet, face mask and self-contained positive pressure or pressure demand breathing apparatus.
- Exposure to excessive heat may lead to venting or rupture of the sealed battery, exposing the internal components which are corrosive and flammable.
- Establish a 20' or larger perimeter if fire or smoke is present at the ESS. PPE must be worn within the perimeter.

SMALL FIRES (not involving the HV batteries)

- Proper PPE gear is required if smoke is present, including a self-contained breathing apparatus. Smoke, regardless of the source, should be assumed to be toxic.
- Extinguish small fires that do not involve the high voltage battery using typical vehicle firefighting procedures.

HEAT OR SMOKE PRESENT AT ESS

- If there is heat or smoke present at the battery enclosure, there is a potential battery fire.
- The smoke is toxic and flammable.
- Do not remove or breach ESS or HV enclosure covers under any circumstances including fire. Doing so might result in severe electrical burns, shocks, or electrocution.
- Use large amounts of water to cool the outside of the enclosure.

BATTERY FIRES

- A battery fire can take up to 24 hours to burn out.
- Use large amounts of water to extinguish battery fires. Establish an additional water source as required. Use a thermal imaging device to ensure all heat sources are extinguished. A battery fire has the potential to re-ignite. Use thermal imaging device to ensure that the high voltage battery is completely cooled before towing or moving the vehicle. The battery must be monitored for at least one hour after it is found to be completely cooled.

- Smoke or steam indicates that the battery is still heating. Do not release the vehicle to second responders, such as the law enforcement and towing personnel, until there has been no heating detected for one hour. Always advise second responders that there is a risk of battery re-ignition.

ESS FIRE DETECTION

A battery fire is an extremely unlikely event. To prevent a fire from starting, the ESS system is designed to prevent thermal events from occurring inside the battery packs and warns the operator of a fire inside the battery enclosures. The ESS system uses abuse tested battery pack enclosures. They are ruggedized against external fuel fire exposure, hard short circuits, internal flooding, and over-heating. Each battery has a cooling plate separating each cell to manage cell temperature. The battery's voltage, current, state of health and temperature is actively monitored. If a battery pack is out of limits, the system automatically disconnects the damaged string connecting the faulty battery pack. In case of an enclosure fire, the fire detection system uses temperature sensors to warn of emerging thermal events, initiate a stop system, and display an instrument panel warning. The fire detectors are mounted in each of the ESS enclosures.

FIRE SUPPRESSION

The vehicle is equipped with a fire suppression system. There are fire detection sensors and suppression nozzles in the fuel cell compartment and an extinguisher cylinder located in a streetside light panel. An alarm panel and manual fire suppression activation switch is located in the driver's overhead area of the vehicle.

PRESSURE RELIEF DEVICES (PRD)

Temperature activated Relief Devices (PRD) relieve hydrogen system pressure when temperature rise due to fires. When the temperature rises above 230°F (110°C), the PRD's open - and remain open - to vent escaping gas away from the tanks.

HYDROGEN LEAK

In the event a hydrogen leak is detected, move all passengers away to at least 300 feet (92m) from the vehicle. DO NOT allow anyone to smoke or operate any devices that may create a spark within 100 feet (30m) of the vehicle. Hydrogen fires can burn with invisible flames.

GAS DETECTION

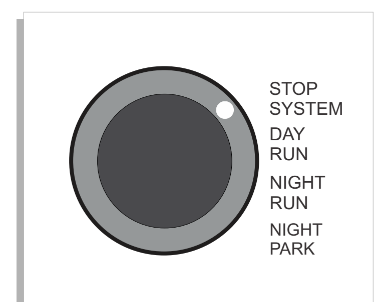
The vehicle hydrogen gas detection system has hydrogen detectors in the hydrogen tank enclosures on the roof, in the fuel cell compartment, and rear interior cabin. A Hydrogen Detection alarm panel is located in the driver's overhead area of the vehicle.

7. IN CASE OF SUBMERSION

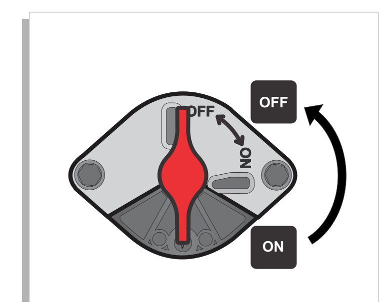
⚠ DANGER

In the event of submersion (if water is above the bottom of the rear bumper), proper PPE gear with a minimum arc flash rating of $8cal/cm^2$ is required before approaching the rear bumper, or ESS compartment area. If the ESS is submerged or partially submerged, a voltage hazard exists at the batteries and may exist between HV energized components. Avoid contact with HV components, cabling, or service disconnects on a submerged vehicle. The front and sides of the vehicle do not pose a HV risk if partially submerged. The vehicle should be retrieved from the water before other work is performed. Water levels below the bottom of the bumper should not pose any HV risk. In some instances, small bubbles may be seen coming from an immersed HV battery. This is referred to as micro-bubbling. This DOES NOT indicate a shock hazard and DOES NOT energize the surrounding water.

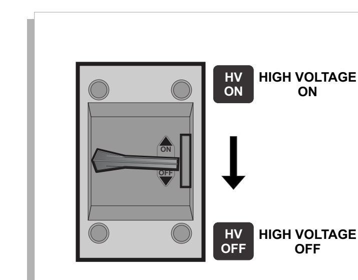
1. Assess vehicle for risks.
2. Shutdown the vehicle by placing the Master Run switch on the driver's side console to the STOP-SYSTEM position.
3. If accessible, turn the 12/24 VDC Battery Disconnect & High Voltage Intelock switches, located behind the rear curbside access door to the OFF position.
4. Avoid contact with High Voltage (HV) components.
5. Attend to any first aid needs.



MASTER RUN SWITCH



BATTERY DISCONNECT SWITCH



HIGH VOLTAGE INTERLOCK SWITCH

⚠ DANGER

High Voltage Cables are routed on the roof, rear compartment and under the vehicle. If these cables become damaged or exposed during an accident, they may remain live. Ensure the High Voltage system is disabled using the High Voltage Interlock switch before working in the area of damaged cables.

6. Access to occupants can be gained through the entrance and exit doors, side windows, or the roof hatches.

8. TOWING / TRANSPORTATION / STORAGE

Proper PPE should be worn when assessing damage around battery enclosures and HV components. If there is damage to an ESS enclosure or battery module, an assessment needs to be made to determine if the vehicle can be safely towed.

DANGER

The batteries are still holding a charge and HV is present at each module.

With the HV system disconnected, the HV cables should not pose a risk during towing. Properly HV trained personnel in proper PPE gear can remove the MSD's at each string to further ensure isolation of the battery packs if component damage is severe and there is concern the automatic disconnects are damaged.Failure to comply with the safety precautions could result in personal injury or vehicle damage. ALWAYS follow the recommended procedures.

The vehicle can be towed from the front using either the flat or raised method. New Flyer recommends the flat towing method to minimize the likelihood of damage to the vehicle. Extra care must be taken when using the raised towing method to ensure adequate ground clearance at the rear of the vehicle. Rear towing is not recommend due to insufficient ground clearance at the front of the vehicle and the problem of locking the front wheels in a straight position.

The operator of the towing vehicle is ultimately responsible for safely securing and towing the vehicle. Ensure that the operator of the towing vehicle is aware of the safety requirements and towing procedures contained in this section. Consult your local Transit Authority for any specific towing procedures and use them carefully in conjunction with the recommended towing procedures contained within this section.

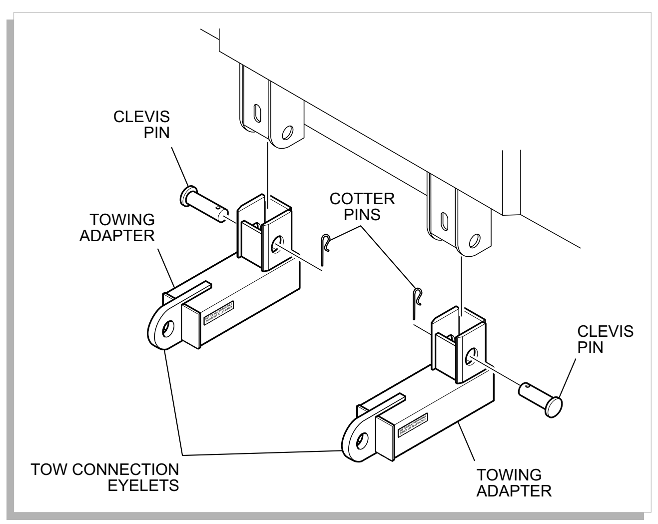
TOWING SAFETY

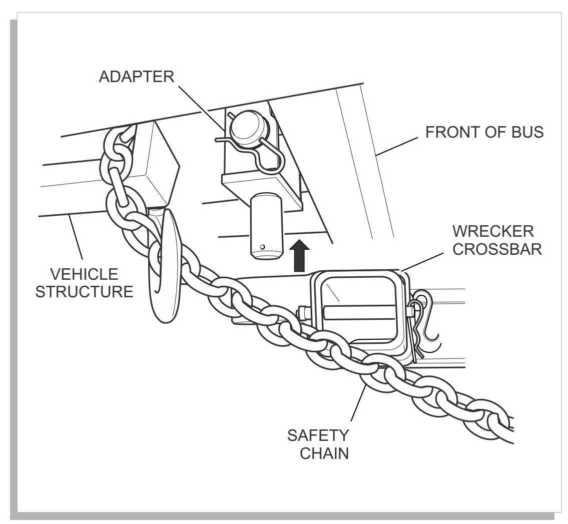
- Follow all State (provincial in Canada) and local traffic regulations.
- All vehicle safety restraint system must be used that is independent of the primary lifting and towing attachments.
- All loose or protruding parts of a damaged vehicle should be secured prior to towing.
- Do not go under a vehicle which is being lifted by the towing equipment, unless the vehicle is adequately supported by safety stands or appropriate blocking.
- No towing operation should be attempted for any reason which jeopardizes the safety of the operator, wrecker, bystanders or other motorists.

- Do not exceed the recommended maximum speed of 35 mph (55 km/h) while towing. Reduce speed over uneven roads, railway tracks or other obstacles.
- Do not exceed the maximum front and minimum rear clearance specifications when the vehicle is raised.
- The vehicle being towed must have its steering secured with the wheels positioned straight ahead.
- If the vehicle being towed is not equipped with an electrical plug for operating the vehicle tail lights, a light bar must be placed at the rear bumper of the towed vehicle.
- Vehicles with an articulated joint cannot exceed a maximum vertical joint angle of 10°. Damage to the articulating joint may occur if this angle is exceeded.
- Switch the Battery Disconnect and the High Voltage Interlock switches, located on the fusebox, to the OFF position.

FLAT TOWING

1. Prepare the vehicle for towing by removing either the driveshaft or both rear axle shafts. Refer to procedures in this document.
2. Obtain an approved towing adapter kit if one is not already provided. The towing adapter used for flat towing consists of two L-shaped brackets, clevis pins and cotter pins.
3. Install each tow adapter into a receiver and locate with a clevis pin.
4. Secure each clevis pin with a cotter pin.
5. Attach the towing vehicle equipment to the tow eye of each towing adapter. The method used will vary depending on the type of towing equipment available.
6. Secure the towing vehicle to the tow adapters. The method used will vary depending on the type of towing equipment available.
7. Attach two safety restraint chains from the towing vehicle to a fixed location on the towed vehicle.
8. Connect the towing vehicle air line, electrical harness (if applicable) and any other safety equipment as required by local regulations to the towed vehicle.

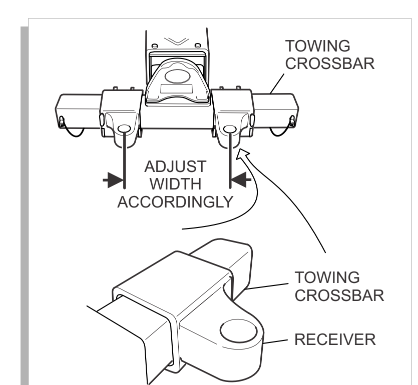


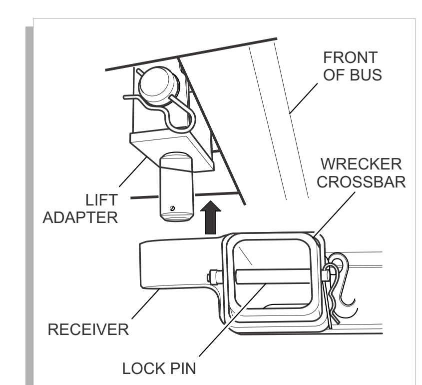


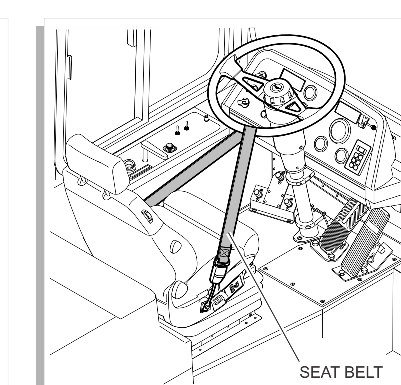
RAISED TOWING

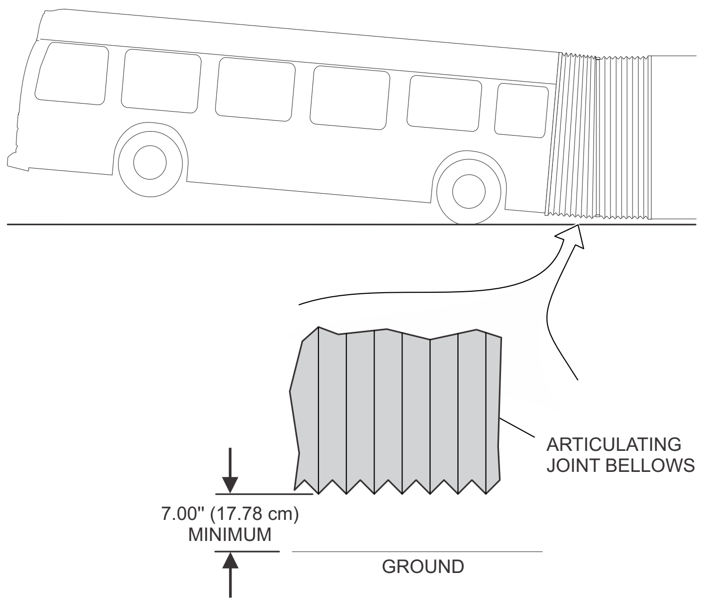
1. Prepare the vehicle for towing by removing either the driveshaft or both rear axle shafts. Refer to procedures in this document.
2. Obtain an approved towing adapter kit if one is not already provided. The towing adapter kit used for raised towing is a peg and socket configuration that consists of two U-shaped lift adapters that attach to the towed vehicle and two lift receivers that slide onto the towing vehicle crossbar.

3. Install the lift adapters onto the towed vehicle as follows:
   a. Slide lift adapter into vehicle receiver and locate with a clevis pin.
   b. Secure each clevis pin with a cotter pin.
4. Install the lift receivers onto the towing vehicle crossbar and slide into position so that they align with the towed vehicle lift adapters.
5. Position the wrecker's lifting boom with the lift receivers in place under the pegs of the lift adapters.
6. Slowly raise the boom until the socket on the lift receivers engage the pegs on the lift adapters. Make any necessary adjustments to the lift receiver positions to ensure proper engagement.
7. Continue to raise the lifting boom until the lift adapters are fully engaged into the receivers.
8. Insert the lock pin through the lift adapters.
9. Raise the front wheels to the height recommended for the specific vehicle being towed and check vehicle ground clearance.
10. Connect the towing vehicle air line, electrical harness (if applicable) and any other safety equipment as required by local regulations to the towed vehicle.
11. Attach two safety restraint chains from the towing vehicle to a fixed location on the towed vehicle.
12. Check to ensure that all clevis and cotter pins are properly inserted, towing equipment is fully engaged and safety chains are clear of the vehicle body before final raising and towing the vehicle.
13. Secure the steering system as follows:
    a. Rotate the steering wheel to position the wheels in the straight ahead position.
    b. Secure the steering system in this position by looping the driver's seat belt around the lower portion of the steering wheel and clipping it into the seat belt buckle.









MAXIMUM HEIGHT

Maximum raised height, as measured from the bottom of the front tire to the ground must not exceed 9.0 inches (22.8 cm).

MAXIMUM VEHICLE GROUND CLEARANCE

Rear bellows clearance must not be less than 7.0 inches (17.78 cm) as measured from the bottom of the articulating joint bellows to the ground.

DRIVESHAFT REMOVAL

This procedure describes removing the driveshaft from the interior of the vehicle which eliminates jacking the vehicle and placing it onto supports.

1. Check that the vehicle is on solid ground and is safe to enter.
2. Place safety triangles on the road to warn approaching vehicles.
3. Remove the driveshaft access door and driveshaft plate.
4. Remove the driveshaft and pull it through the access opening.
5. Reinstall the access plate to keep debris out of vehicle.
6. Setup the towing equipment and retrieve the safety triangles.

REAR AXLE SHAFT REMOVAL

The axle shafts or the driveshaft must be removed before towing the vehicle.

1. Remove the ten M6 bolts that retain the axle shaft to the hub and carefully slide out the axle shaft.
2. Seal the axle housing opening using a commercially available cover plate.
3. Protect the axle shaft and temporarily store inside the vehicle while being towed.
4. Repeat procedure for opposite axle shaft.
5. Vehicle is now ready for towing.

SPRING BRAKE RELEASE

The following procedure is required to mechanically release the emergency/parking spring brake during towing if the brakes cannot otherwise be released using air pressure. The brake chamber is considered to be “caged” when the spring is compressed sufficiently to release the emergency/parking brake. Rotating the release bolt counterclockwise will compress the spring.

1. Locate the release bolt in the center of the brake chamber.
2. Cage the rear brake chamber by unscrewing the release bolt. Do not exceed a force of 74 ft-lb. (100 Nm) during caging. The chamber is caged when the head of the release bolt extends approximately 1.36 ± 0.12" beyond the surface of the brake chamber.

VEHICLE REMOVAL FROM DITCH

PULLING FROM THE FRONT

1. Connect a chain hook onto one side of each tow adapter pin hole, in the front structural tow connectors, located behind the front bumper.
2. Protect the front bumper during pulling with a 4" x 4" or equivalent wood block placed between the chains and the bottom of the front bumper.
3. With the assistance of a driver to steer the vehicle, release the parking brake and pull the vehicle from the ditch.

PULLING FROM THE BACK

1. Connect a chain hook onto each tow pin located in each main rail behind the rear bumper.
2. Protect the rear bumper during pulling with a 4" x 4" or equivalent wood block placed between the chains and the bottom of the rear bumper.
3. With the assistance of a driver to steer the vehicle, release the parking brake and pull the vehicle from the ditch.

9. IMPORTANT ADDITIONAL INFORMATION

IN THE EVENT OF A BATTERY SPILL

CAUTION

The Electrolyte and its fumes are toxic. Wear full protective clothing, including helmet and face mask, and self-contained positive pressure or pressure demand breathing apparatus. If possible, stop or minimize the flow of battery electrolyte. Contain spillage from a damaged or open cell or battery with dry sand or an absorbent cloth. Battery Electrolyte is classified as a water pollutant under the Clean Water Act and should be prevented from contaminating soil or from entering sewage and drainage systems which lead to waterways.

As an immediate precautionary measure, isolate spill or leak area for at least 75 feet (25m) in all directions. Keep unauthorized personnel away. Stay upwind. Keep out of low areas. Ventilate closed areas before entering.

EXPOSURE HAZARDS

- **Eyes - Not dangerous with normal use. Eye contact with contents of an open battery can cause severe irritation or burns to the eye.**
- **Skin - Not dangerous with normal use. Skin contact with contents of an open battery can cause severe irritation or burns to the skin.**
- **Inhalation - Inhalation of materials from a sealed battery is not an expected route of exposure. Material emitted from a ruptured battery may cause respiratory irritation.**
- **Ingestion - Swallowing of materials from a sealed battery is not an expected route of exposure. Swallowing the contents of an open battery can cause serious chemical burns of mouth, esophagus, and gastrointestinal tract.**

FIRST AID MEASURES

- **Eye contact - If eye comes in contact with contents of an open or damaged cell or battery, immediately flush the contaminated eye(s) with lukewarm water for at least 30 minutes. Get medical attention immediately. Rinse eye with calcium gluconate solution (1%) until arrival of doctor. Continue rinsing.**
- **Skin contact - If skin contact with contents of an open battery occurs, immediately flush with lukewarm water for at least 30 minutes. Thoroughly wash (or discard) clothing and shoes before reuse.**

- **Inhalation - If contents of an opened battery are inhaled, remove source of contamination or move victim to fresh air and seek medical advice.**
- **Ingestion - If ingestion of contents of an open battery occurs, rinse mouth thoroughly with water. DO NOT induce vomiting. If victim is fully conscious, give a cupful of water. Never give anything by mouth to an unconscious person. If vomiting occurs, keep head lower than the hips to help prevent aspiration. Call a physician or poison control center immediately.**

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